

Boughton Island

Shoreline length: 17 km



Shore Types

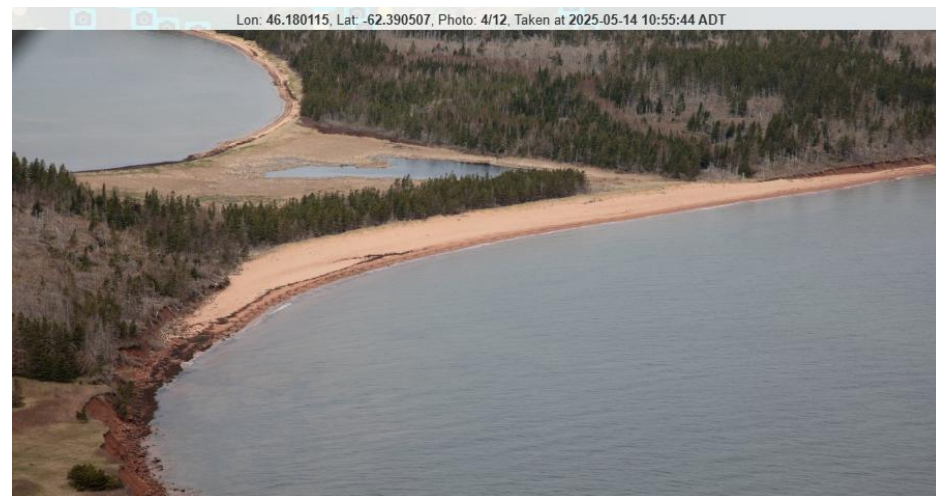
Boughton Island is at the entrance of Cardigan estuary. It is connected to the mainland by a barrier beach. Main shore types are sand dunes and low plains.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
22	10	0	15	17	0	36	0



North end of barrier dune to Boughton Island - Source: CHRIS



South cliffs and barrier dune on Boughton Island - Source: CHRIS

Governance, Infrastructure, and Economy

Boughton Island is almost entirely unincorporated; roughly 2% falls within the Town of Three Rivers. The island is owned by the Government of Prince Edward Island. The island is accessed only through a single point connecting to Bruce Point Road; all roads are unpaved, and this sole access point is a flood hazard risk. There is no alternate land access. There are no coastal economic drivers and no legacy development constraints noted.

Natural Assets and Heritage

Boughton Island is protected under the Natural Areas Protection Act and is home to the Boughton Island Nature Preserve. It is frequently visited by hikers. The shoreline includes critical habitat for bird species including the endangered Piping Plover and threatened Bank Swallow.

Planning and Policy Framework

Provincial rules apply except for the small stretch of shore located within Three Rivers. Under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

Coastal Flood and Erosion Risk

There is no infrastructure on the barrier or island however there are a few buildings behind the proximal point of the sandpsit on the

Proposed SMP Strategies

The primary strategy for Boughton Island is No Active Intervention. In the only area where the few buildings would be at long-term risk, managed realignment should be considered. It is possible that the barrier beach may migrate inland overtime. Maintaining natural sediment supply will be critical to minimize the risk of breaches.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	NAI	NAI	NAI
Rationale	The local shoreline and hazard zone should remain natural, undeveloped.		

mainland. Those buildings located behind sand dunes and wetlands are vulnerable to long-term flood hazards. Erosion hazard is highest along the undeveloped sand dunes and wetlands. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential shoreline change or salt marsh migration to future time horizons.

Number of at-risk buildings [N. per km shoreline]

	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
SMU Boughton Island				
Flood risk	0	0	0	1 [<1/km]
Erosion risk	N/A	1 [<1/km]	1 [<1/km]	1 [<1/km]

Sediment transport - Sediment transport is highest along the exposed sand dunes and wetlands and feeds the barrier beaches.



LEGEND - Shoreline Management Unit - Net Longshore Sediment Transport - High Hazard Area - Flood Extent - Present Day (100-year Return) - Flood Extent - 1.2m SLR (100-year Return) - Property Boundary - Building > 90ft²	0 110 220 440 Metres Coordinate System: NAD 83 CSRS PEI Double Stereographic NOTES:		COASTAL EROSION AND FLOOD HAZARD Shoreline Management Unit: Boughton Island	<i>Prince Edward Island</i> Île-du-Prince-Édouard CBCL
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